Runbook: Orchestrated Pipeline Step Failure (Step Functions)

Runbook ID: RB_PIPELINE_STEP_FAILURE_DOC

Symptom

The nemoguard-daily-pipeline Step Functions execution enters a FAILED status. One of its two states (IngestCustomerProfile or ProcessOrderEvents) raised an unhandled exception.

Likely Root Cause

The root cause is whichever Lambda the FAILED state actually invoked, NOT the orchestration layer itself. Step Functions is just the wrapper -- diagnose the underlying Lambda's real failure using the same runbooks as its standalone counterpart.

Diagnostic Steps

1. Call describe_step_function_execution with the execution ARN to see exactly which state failed and the failure event details.
2. If IngestCustomerProfile failed: follow RB_SCHEMA_DRIFT_DOC.
3. If ProcessOrderEvents failed: follow RB_PARTIAL_WRITE_DOC.
4. Because ProcessOrderEvents only runs AFTER IngestCustomerProfile succeeds, if IngestCustomerProfile failed, ProcessOrderEvents never ran at all -- do not treat this as a partial-write scenario.

Remediation Steps

1. Fix the root-cause Lambda per its own runbook.
2. Re-invoke the ENTIRE state machine execution from the start (StartAt), not just the failed state -- Step Functions standard executions do not support resuming from a specific state.
3. If IngestCustomerProfile succeeded before ProcessOrderEvents failed, re-running the whole execution will re-invoke IngestCustomerProfile too -- confirm this is idempotent (it is, for this lab's ingest job) before doing so, or split remediation into a manual two-step process if not.

Verification

Call describe_step_function_execution on the NEW execution ARN and confirm status is SUCCEEDED with no recent_failure_events.

Risk Classification

MEDIUM. Re-running an entire multi-step execution can have side effects beyond the failed step if upstream steps are not verified idempotent.